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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (EVT) (CBCS) Degree Examinations

May-June 2025

22EVT202 – EV MOTOR DRIVES AND CONTROL

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	What are the major requirements of EV propulsion system?	6	L*1	1	3,5
b)	Explain the concept of field weakening in DC motors.	6	L2	1	3,5
c)	A DC motor has following parameters: Voltage(rated)=220V, rated armature current = 16A, rated speed = 1220 rpm, Torque(rated) = 26 Nm, Resistance $R_a = 0.8 \Omega$. Determine				
i.	Back EMF				
ii.	Back EMF constant				
iii.	Torque constant				
iv.	Motor speed when armature current is 10A.	8	L2	1	3,5
2. a)	Explain the desired Speed Vs Torque characteristics of EV drive system. Specify and explain constant torque and constant power regions of operation.	6	L2	1	3,5
b)	Develop the DC motor electrical and mechanical model with the help of DC motor equivalent circuit. Also explain emf and Torque generation phenomenon in DC motors.	8	L3	1	3,5
c)	Illustrate four quadrant operation of a DC motor.	6	L2	1	3,5
Unit – II					
3. a)	Consider a DC motor with $R_a = 0.4 \Omega$, $L_a = 500 \mu H$, $K_t = 0.05 \text{ Nm/A}$, $J = 2 \times 10^{-4} \text{ kg} \cdot \text{m}^2$, and $B = 1 \times 10^{-5} \text{ Nm} \cdot \text{s}$.				
a)	Determine current PI gains, K_{pc} and K_{ic} so that the current control bandwidth is 200 Hz and the damping coefficient is 0.707.				
b)	Treating the current block as unity, determine speed loop PI gains, K_{ps} and K_{is} so that the speed loop bandwidth is 10 Hz and damping coefficient is 1.	12	L3	2	3,5
b)	Summarize PD controller and its use cases along with bode plot.	8	L2	2	3,5
4. a)	Explain PI controller with reference model with the help of block diagram.	8	L2	2	3,5
b)	Outline the objective of controller in any system. Define sensitivity function and complementary sensitivity function.	6	L2	2	3,5
c)	With the help of block diagram, explain 2 DOF control system.	6	L2	2	3,5
Unit – III					
5. a)	List the differences between induction motor and 3-phase transformer.	6	L1	3	3,5
b)	Summarize the third order harmonic injection technique in SPWM generation scheme.	6	L2	3	3,5
c)	Explain VVVF control of IM along with speed-torque characteristics curve.	8	L2	3	3,5
6. a)	Illustrate Field Oriented Control implementation of IM in rotor reference frame with the help of a block diagram.	12	L3	3	3,5
b)	Explain line starting of Induction Motor. Also mention the problems with line starting and potential solution.	8	L3	3	3,5

P.T.O

Unit – IV

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| 7. a) | Explain the control block diagram of PMSM using the coordinate transformation map. | 8 | L2 | 4 | 3,5 |
| b) | Summarize PMSM voltage and current limits. | 6 | L3 | 4 | 3,5 |
| c) | Illustrate how the machine sizing varies under same power based on maximum torque and speed. | 6 | L2 | 4 | 3,5 |
| 8. a) | Elaborate the torque generation in PMSM. | 8 | L2 | 4 | 3,5 |
| b) | List the types of PMSM along with cross section diagram. | 6 | L1 | 4 | 3,5 |
| c) | Explain Maximum Torque per Ampere Control in PMSM. | 6 | L2 | 4 | 3,5 |

Unit – V

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| 9. a) | How hall effect is used to measure current? Explain. | 6 | L2 | 5 | 3,5 |
| b) | Illustrate the working of Encoders and Hall Position Sensors. | 8 | L2 | 5 | 3,5 |
| c) | Write a brief note on resolver and R/D converter. | 6 | L3 | 5 | 3,5 |
| 10. a) | Analyze the significance of current sampling time in motor control inverters. | 6 | L4 | 5 | 3,5 |
| b) | Summarize on Current Model Estimator for IM. | 8 | L2 | 5 | 3,5 |
| c) | What is the importance of position and current sensing in an EV motor control? Explain. | 6 | L2 | 5 | 3,5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
